

ECONOMETRIC THEORY AND PRACTICE I

Quiz 3 - Regression

3 questions | 10 minutes | 1 point each | No negative marking



Keep the QR code and corner marks clear.

form ref: QJ2H

Fill all 3 ID digits clearly. An unassignable ID receives zero.

STUDENT ID - 3 DIGITS

Write each digit, then fill its bubble.

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RESPONSE PANEL

Fill exactly one bubble in each row.

Question 1	☐ A	☐ B	☐ C	☐ D	☐ E	☐ F	☐ G
Question 2	☐ A	☐ B	☐ C	☐ D	☐ E	☐ F	☐ G
Question 3	☐ A	☐ B	☐ C	☐ D	☐ E	☐ F	☐ G

1. The fitted OLS line is predicted $Y = 100 - 2X$. One observation has $X=10$ and $Y=83$. What is its OLS residual?

- | | |
|--------|--------|
| A. -17 | B. -37 |
| C. 80 | D. 3 |
| E. -3 | F. 163 |
| G. 83 | |

2. An OLS regression with an intercept and one regressor uses $n=27$ observations. $SER=2$ and $R^2=0.75$. SSR denotes the sum of squared residuals; ESS denotes explained sum of squares. Which pair gives (SSR ; ESS)?

- | | |
|---------------|-----------------|
| A. (300; 100) | B. (100; 100/3) |
| C. (100; 400) | D. (108; 324) |
| E. (50; 150) | F. (100; 75) |
| G. (100; 300) | |

3. In the constant-effect causal model $Y=\alpha+\beta X+u$, $X=10$ with probability $1/4$ and $X=14$ with probability $3/4$. $E(u)=0$, $E(u | X=10)=3$, and $E(Y | X=14)-E(Y | X=10)=8$. Which pair gives (β ; $E(u | X=14)$)?

- | | |
|--------------|-------------|
| A. (1; -1) | B. (2; -1) |
| C. (3; -1) | D. (12; -1) |
| E. (3; 1) | F. (2; 0) |
| G. (3.5; -3) | |

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Question 2	☐ A	☐ B	☐ C	☐ D	☐ E	☐ F	☐ G
Question 3	☐ A	☐ B	☐ C	☐ D	☐ E	☐ F	☐ G

1. The fitted OLS line is predicted $Y = 110 - 3X$. One observation has $X=10$ and $Y=84$. What is its OLS residual?

- | | |
|--------|--------|
| A. -4 | B. 164 |
| C. -56 | D. 84 |
| E. 80 | F. -26 |
| G. 4 | |

2. An OLS regression with an intercept and one regressor uses $n=27$ observations. $SER=3$ and $R^2=0.75$. SSR denotes the sum of squared residuals; ESS denotes explained sum of squares. Which pair gives $(SSR; ESS)$?

- | | |
|------------------|---------------|
| A. (225; 900) | B. (243; 729) |
| C. (75; 225) | D. (225; 675) |
| E. (225; 75) | F. (675; 225) |
| G. (225; 168.75) | |

3. In the constant-effect causal model $Y = \alpha + \beta X + u$, $X=10$ with probability $1/4$ and $X=14$ with probability $3/4$. $E(u)=0$, $E(u | X=10)=6$, and $E(Y | X=14) - E(Y | X=10)=8$. Which pair gives $(\beta; E(u | X=14))$?

- | | |
|------------|-------------|
| A. (4; -2) | B. (2; -2) |
| C. (0; -2) | D. (16; -2) |
| E. (4; 2) | F. (2; 0) |
| G. (5; -6) | |

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form ref: QKMC

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Question 3	☐ A	☐ B	☐ C	☐ D	☐ E	☐ F	☐ G

1. The fitted OLS line is predicted $Y = 120 - 4X$. One observation has $X=10$ and $Y=85$. What is its OLS residual?

- | | |
|--------|--------|
| A. 5 | B. 85 |
| C. 165 | D. -35 |
| E. -5 | F. 80 |
| G. -75 | |

2. An OLS regression with an intercept and one regressor uses $n=27$ observations. $SER=4$ and $R^2=0.75$. SSR denotes the sum of squared residuals; ESS denotes explained sum of squares. Which pair gives $(SSR; ESS)$?

- | | |
|----------------|-----------------|
| A. (400; 300) | B. (1200; 400) |
| C. (100; 300) | D. (400; 400/3) |
| E. (400; 1200) | F. (432; 1296) |
| G. (400; 1600) | |

3. In the constant-effect causal model $Y=\alpha+\beta X+u$, $X=10$ with probability $1/4$ and $X=14$ with probability $3/4$. $E(u)=0$, $E(u | X=10)=9$, and $E(Y | X=14)-E(Y | X=10)=8$. Which pair gives $(\beta; E(u | X=14))$?

- | | |
|-------------|--------------|
| A. (20; -3) | B. (5; -3) |
| C. (5; 3) | D. (6.5; -9) |
| E. (2; 0) | F. (2; -3) |
| G. (-1; -3) | |

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Question 3	☐ A	☐ B	☐ C	☐ D	☐ E	☐ F	☐ G

1. The fitted OLS line is predicted $Y = 130 - 5X$. One observation has $X=10$ and $Y=86$. What is its OLS residual?

- | | |
|--------|--------|
| A. -6 | B. -44 |
| C. 166 | D. -94 |
| E. 6 | F. 80 |
| G. 86 | |

2. An OLS regression with an intercept and one regressor uses $n=27$ observations. $SER=5$ and $R^2=0.75$. SSR denotes the sum of squared residuals; ESS denotes explained sum of squares. Which pair gives (SSR ; ESS)?

- | | |
|------------------|-----------------|
| A. (625; 468.75) | B. (625; 1875) |
| C. (125; 375) | D. (625; 625/3) |
| E. (1875; 625) | F. (675; 2025) |
| G. (625; 2500) | |

3. In the constant-effect causal model $Y=\alpha+\beta X+u$, $X=10$ with probability $1/4$ and $X=14$ with probability $3/4$. $E(u)=0$, $E(u | X=10)=12$, and $E(Y | X=14)-E(Y | X=10)=8$. Which pair gives (β ; $E(u | X=14)$)?

- | | |
|-------------|-------------|
| A. (2; 0) | B. (2; -4) |
| C. (6; 4) | D. (-2; -4) |
| E. (8; -12) | F. (6; -4) |
| G. (24; -4) | |

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RESPONSE PANEL

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Question 2	☐ A	☐ B	☐ C	☐ D	☐ E	☐ F	☐ G
Question 3	☐ A	☐ B	☐ C	☐ D	☐ E	☐ F	☐ G

1. The fitted OLS line is predicted $Y = 140 - 6X$. One observation has $X=10$ and $Y=87$. What is its OLS residual?

- | | |
|--------|---------|
| A. 167 | B. -53 |
| C. 7 | D. 80 |
| E. -7 | F. -113 |
| G. 87 | |

2. An OLS regression with an intercept and one regressor uses $n=27$ observations. $SER=6$ and $R^2=0.75$. SSR denotes the sum of squared residuals; ESS denotes explained sum of squares. Which pair gives (SSR ; ESS)?

- | | |
|----------------|----------------|
| A. (900; 675) | B. (2700; 900) |
| C. (900; 3600) | D. (972; 2916) |
| E. (150; 450) | F. (900; 2700) |
| G. (900; 300) | |

3. In the constant-effect causal model $Y=\alpha+\beta X+u$, $X=10$ with probability $1/4$ and $X=14$ with probability $3/4$. $E(u)=0$, $E(u | X=10)=15$, and $E(Y | X=14)-E(Y | X=10)=8$. Which pair gives (β ; $E(u | X=14)$)?

- | | |
|-------------|---------------|
| A. (-3; -5) | B. (2; -5) |
| C. (28; -5) | D. (7; 5) |
| E. (7; -5) | F. (9.5; -15) |
| G. (2; 0) | |